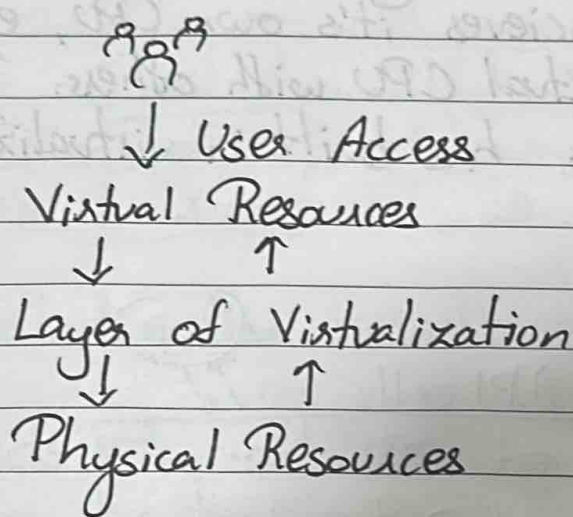


Virtualization:

It is the ability that allows sharing the physical instance of a single application or resource among multiple organizations or users.

It refers to representation of physical computing resources in simulated form having made through the software. This special layer is referred as layer of virtualization.

This layer transforms physical computing resources into virtual form which users use to satisfy their computing needs.



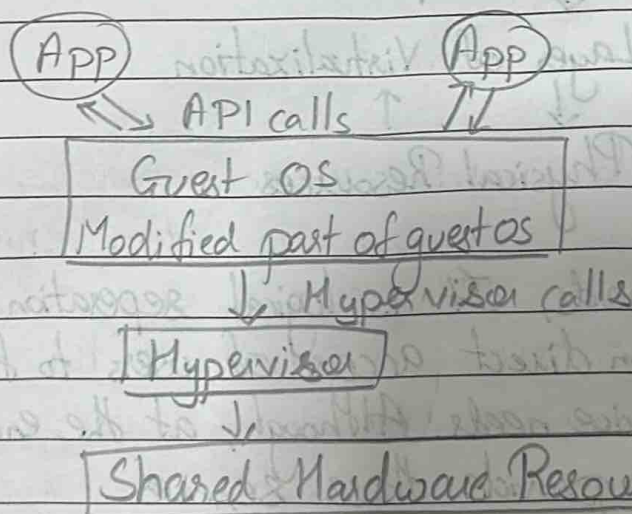
In simple words, It is logical separation of physical resources from direct access of users to fulfill their ~~resources~~ service needs. Although, at the end physical responsible to provide those services.

It provides a level of logical abstraction that liberates user-installed software from being tied to a ~~es~~ hardware. Rather user installs everything on the logical operating ~~env~~ environment created through virtualization.

* Types of Virtualization

1] CPU Virtualization

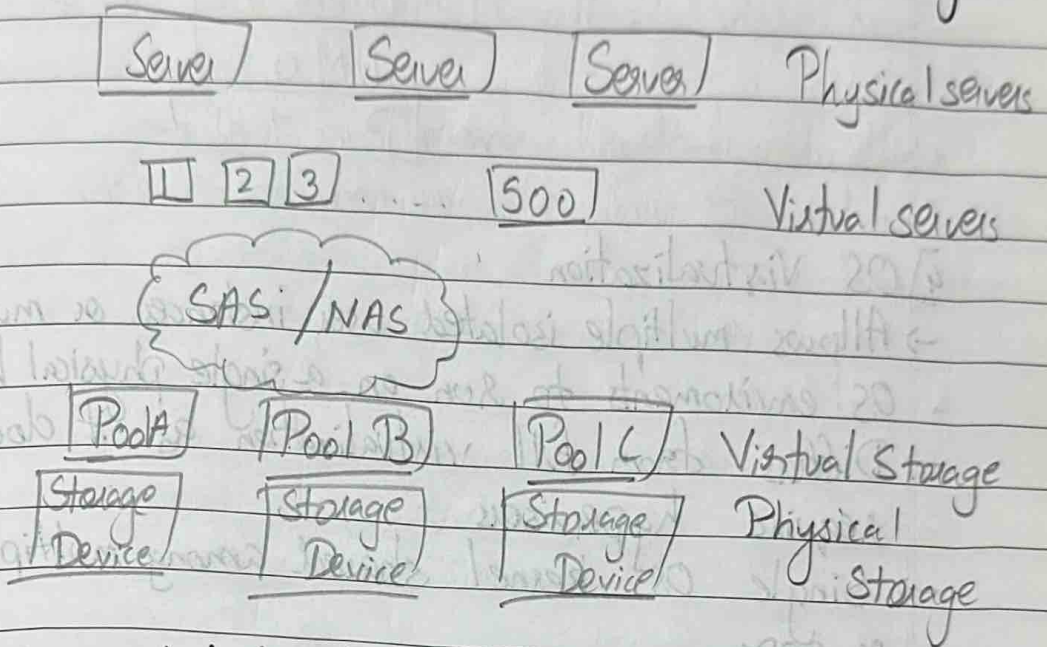
- Allows multiple OS to share a single physical CPU by creating virtual CPUs
- Hypervisor manages CPU access for multiple VM's
- Hypervisors sit b/w physical CPU & VM's managing CPU cycles & ensuring fair distribution among VM's
- Each VM perceives it's own CPU, even though it shares the actual CPU with others.
- Modern CPU's has built in virtualization to improve efficiency



2] Storage Virtualization

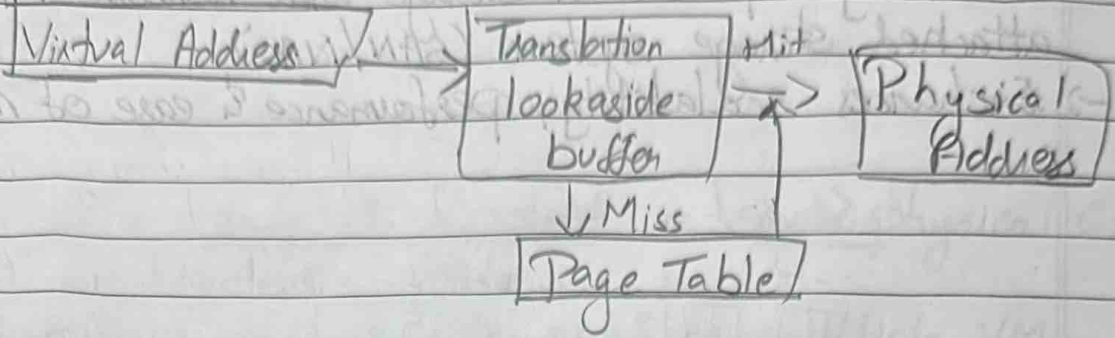
- It combines multiple physical storage devices into single logical storage unit, abstracting underlying hardware
- Virtualization pools multiple storage devices, allowing centralized management & allocation of storage resources.
- Storage appears as single unit to the OS, even if it consists of multiple physical drives spread across diff locations

- Commonly implented in storage area network & network attached storage systems (SAN/NAS)
- Enhances scalability, performance & ease of management.



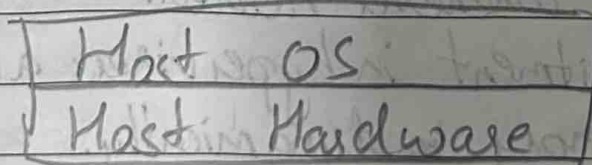
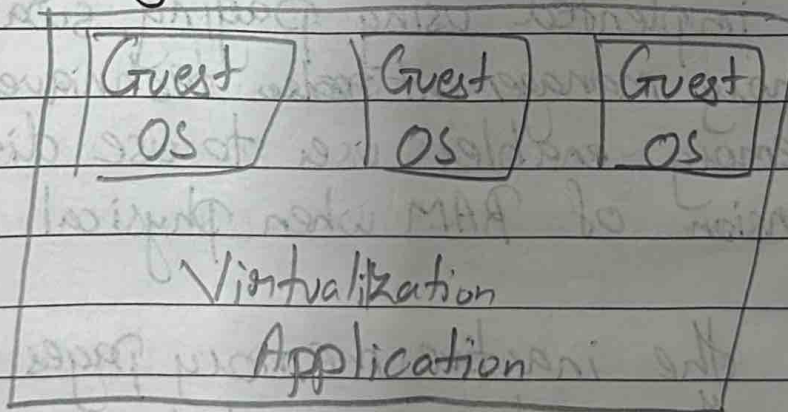
3] Memory Virtualization

- It abstracts physical memory ^(RAM) and provides applications with illusion of having more than physically available
- Commonly implented using paging, swapping and other memory management ~~techniques~~ techniques.
- Virtual memory enables user to use disk storage as an extension of RAM when physical memory is insufficient.
- OS swaps the inactive memory pages to disks and retrieves them whenever needed.
- Memory overcommitment in hypervisors allows multiple VM's to share memory ~~in~~ dynamically.



4) OS Virtualization

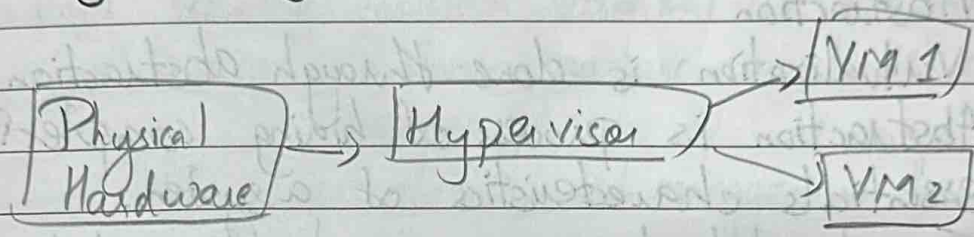
- Allows multiple isolated user instances or multiple OS environments to run on a single physical hardware.
- Differs from full virtualization as it does not require hypervisors.
- Single OS kernel shared among multiple users or apps.
- Unlike traditional VM's which have separate OS instances, containers share same kernel but operate in isolated env.
- Containerization technologies provide lightweight, scalable app deployment.



Eg: Docker.

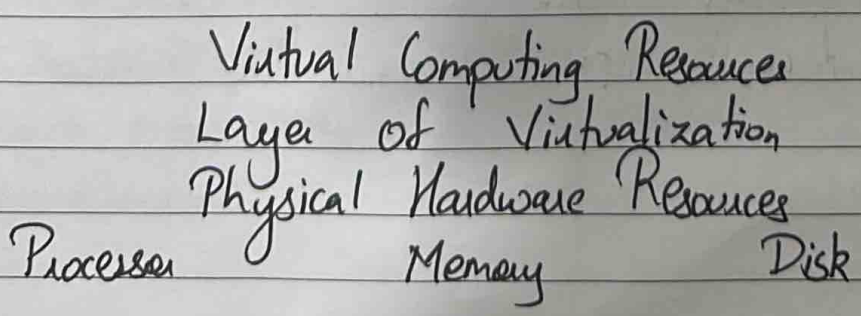
5) Hardware Virtualization

- Allows multiple OS to run on a single physical server by abstracting hardware components.
- Hypervisor manages VM's & allocates virtual hardware resources
- VM's operates as independent systems with their own OS, despite sharing the same physical hardware.
- Type 1 hypervisors run directly on the hardware while Type 2 hypervisors run inside existing OS.



* Virtualization of Physical Computing Resources

- Any kind of computing devices can be virtualized apart from basic computing devices like processor, primary memory
- Almost all kind of computing devices can be virtualized like storage, memory, CPU and more.
- A virtualized component can only be operational when a physical resource empowers it from back end.
- Virtual processor can only work when there is a Phy processor linked with it



- layer of virtualization transforms physical computing devices to virtual form & present to users.
- Users can install OS into VM's just like they do in physical computer.
- OS on VM is known as guest OS.
- When virtualization technique is in place, guest OS executes as if it were running directly on physical machine.

* Abstraction

- Virtualization is done through abstraction.
- Abstraction is process of hiding complex & non-essentials characteristics of a system.
- Through abstraction, sys can be presented in simplified manner for some particular use by omitting unwanted details from users.
- In computing, abstraction is implemented through the layers of software. The layer of OS can be treated as layer of abstraction.
- In CC, resource virtualization which acts as a layer of ~~the~~ s/w over phy resources to create virtual resources acts as a layer of abstraction. This abstraction makes it easier to offer more flexible, scalable & powerful resources.

* Business Benefits of Virtualization

- ① Cost Reduction: Reduces h/w cost of maintenance by consolidating multiple servers into a few powerful machines.
- ② Scalability:- Enables rapid resource allocation & scaling up/down based on demand.
- ③ Improved Disaster Recovery: VM's can be backed up and restored easily.
- ④ Enhanced security: Isolated virtual env. reduce security risk.
- ⑤ Energy efficiency: Reduces power consumption & cooling requirements.
- ⑥ Simplified Management: Centralized management through hypervisors simplifies IT operations.

* Machine / Server Virtualization

- It enables multiple VM's to run on one physical server.
- Phy server is the host while each virtual instance is a guest system.
- Breaks traditional one to one rule where one comp could support only one OS.
- You can now run several OS's on single machine simultaneously.
- Each guest sys operates in isolation, running its own sw & apps.
- These virtual sys securely share & access the underlying phy resources.
- Improves h/w efficiency by maximizing single physical machine.

2.3

* Hypervisors:

- Virtualization requires use of hypervisor, which was initially called VM Monitor or VMM.
- It abstracts OS & apps from their underlying h/w
- Phy h/w where hypervisor runs on is known as host machine, whereas the VM's it creates are known as guest machines.
- Let's host h/w operate multiple VM's independent of each other & share abstracted resources among them
- It helps to increase a data generation efficiency compared to phy workload hosting

1] Type 1 Hypervisors

- Also called bare-metal hypervisors or native hypervisors
- It takes place of host OS
- Runs directly on physical hardware, interacting directly with CPU, phy memory & storage.

→ Adv:

- High performance due to direct h/w access
- Enhanced security with isolation b/w VM's
- Better resource management & scalability

→ Dis adv:

- Requires dedicated h/w
- Complex setup & management

→ Eg: MS Hyper-V, VMware

2] Type # 2 Hypervisors

- Also called embedded or hosted hypervisors
- Doesn't run directly on underlying h/w
- Instead runs on an application in OS.
- Rarely used in server-based env., suitable for individual PC users to run diff OS
- Runs on top of host OS, which manages h/w
- There are additional toolkits that can help in enabling user to cut paste b/w two or access host OS files & folders.

→ Adv:

→ Easy to install & use

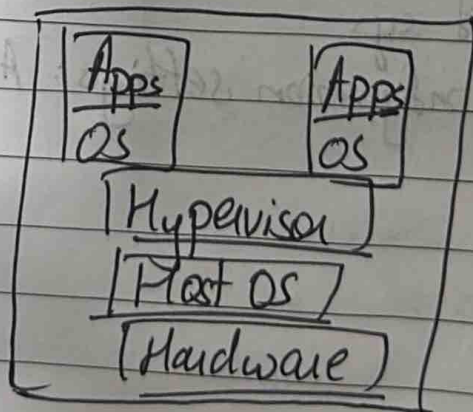
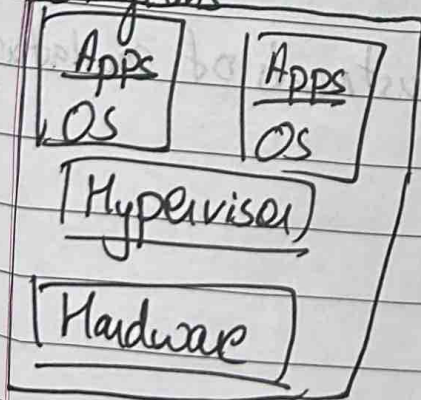
- can run alongside other applications on Host OS
- Ideal for testing & Dev

→ Disadv:

- Lower performance due to reliance on host OS
- Less secure, as it shares resources with host OS.

→ Eg: VMware workstation, Virtual Box.

⇒ Diagrams



Type 1

* High Level Lang VM

- It provides an execution env. for code written in high level programming lang, abstracting h/w details.
- Eg: Java VM, Python VM, .Net Common lang runtime
- Benefits:
 - Portability
 - Performance
 - Security
 - Optimization

* Emulation

- It is capability of a comp program that simulates behavior of another comp sys or prog inside an electronic device
- Capability of program that simulates behavior of another comp sys or program inside an electronic device.
- It allows host sys to run software for guest sys.
- Parts:
 - 1] Emulator Core: Translates instructions of guest sys to that for host sys which it can understand
 - 2] ROMs / BIOS: Contains sys software required to boot and run programs of the emulated sys.
 - 3] User Interface: Allows user to load ROM files & control emulator
 - 4] Graphics & Audio Module: Produces visual & sound output of sys
 - 5] Configuration settings: Allows adjustments of emulator parameters

* Adv of virtualization:

- ① Cost saving - Reduces h/w buying & maintenance cost
- ② Improved scalability - Easily scales up or down based on demand
- ③ Disaster & Recovery Backup - VM's can be restored quickly
- ④ Enhanced Security: Isolates env's to prevent sys
- ⑤ Better Resource Utilization: Allows multiple VM's to share resources

* Disadv

- ① Performance Overhead - Virtualization layers add some latency
- ② Complexity: Requires expertise in config & management
- ③ Security Risks: Misconfig or hypervisor vulnerabilities can leak data
- ④ Hardware limitation: Requires sufficient computing power for smooth operations.

2.3

* Virtualization security threats:

- ① Single Point of Failure: Since one phy host runs on ~~at~~ multiple VM's, one security breach can cause all the sys to crash at once.
- ② Hypervisor Attacks: hypervisor is foundation of virtualized env. If it compromises security of every machine it manages is at risk.
- ③ Complex Configuration: Virtualization adds layers of abstraction that increase sys complexity. If these layers are not setup perfectly, it creates hidden security gaps.

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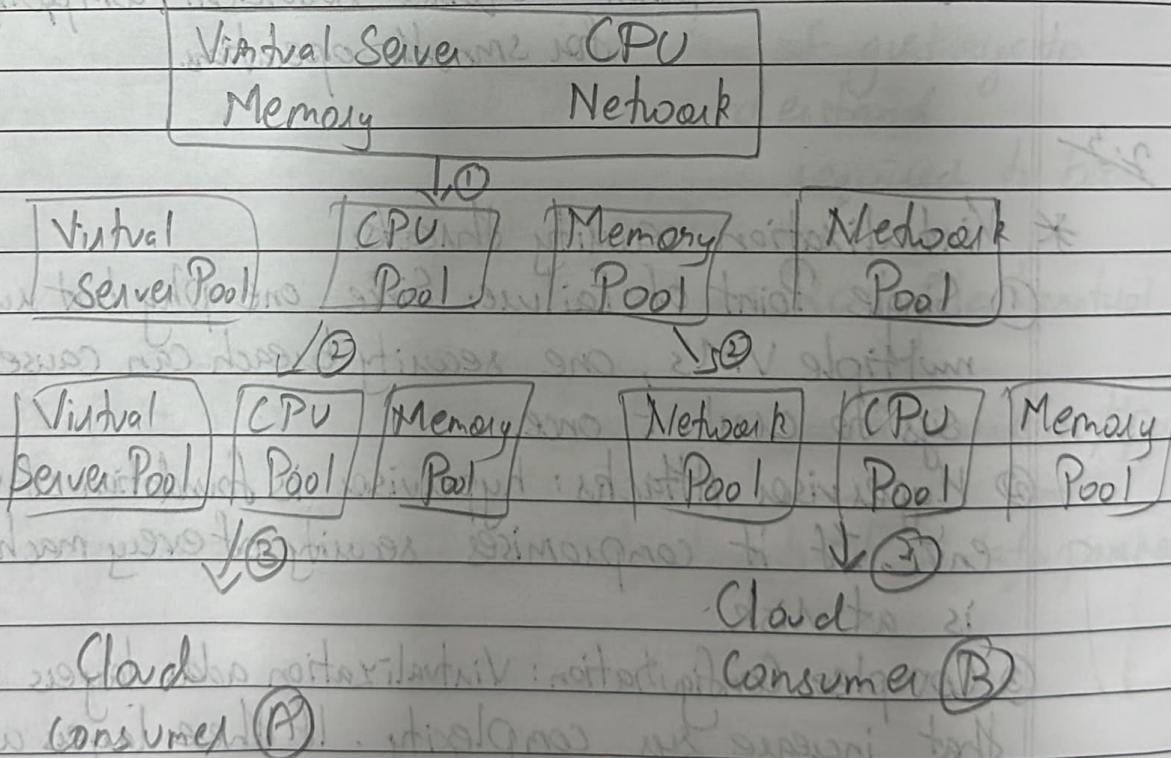
④ Privilege Escalation: Due to design flaws, a user or an app might gain more access to resources than they should have, potentially allowing hacker to control sys. through hypervisor.

⑤ Trust Zone Consolidation: When ^{sys} users with diff levels of security (trust zones) ~~and~~ are combined on same h/w without strict separation, sensitive or highly trusted ~~are~~ apps may become vulnerable to threats from less secure ones.

2.4

* Resource Poding

→ Process of grouping phy computing resources like CPU, memory, etc & dynamically allocating them to multiple VM's or users based on demand.



Eg: In IaaS platforms like AWS, google cloud.

→ Key Aspects:

- ① Logical Resource Allocation: Physical resources are abstracted & managed as an unified pool
- ② Dynamic Allocation: Resources are assigned as needed to VM's & apps
- ③ Centralized Management: Administrators can manage & control resource effectively & efficiently

* Resource Sharing

- Refers to multiple VM's, users or apps accessing & utilizing same physical hardware or software resources simultaneously
- Eg: Type 1 & 2 hypervisors allow multiple VM's to share single phy server.
- Key Aspects:

- ① Multi Tenancy - In cloud envs, multiple tenants share ~~only~~ underlying infrastructure security
- ② Concurrent access - Multiple ~~users~~ VM's share CPU cycle, RAM, storage & network resource

* Resource Provisioning

- Process of allocating computing resource to VM's apps or user based on their requirements.
- Ensures that sufficient resources are available for efficient performance.
- Types:

- ① Static: Fixed allocation of resources, predefined less flexible but ensures consistent performance
- ② Dynamic: Resources are allocated & adjusted dynamically based on workload, changes prevent resource wastage & optimize performance.